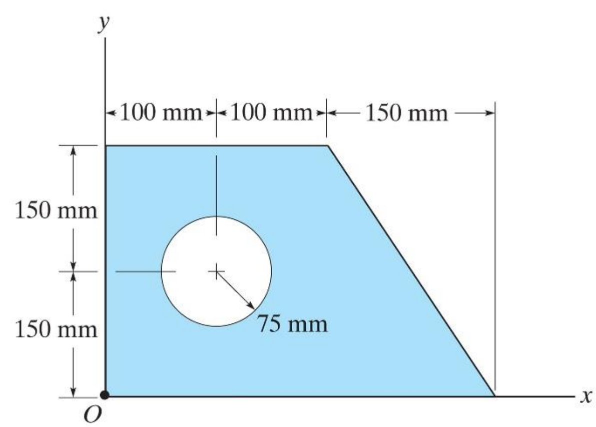
Practice Exam 4 POI Conceptual Questions Review

Section 3 – Product of Inertia

POI Concepts:

1. What are units of MOI? POI?
2. How do areas (A) and distance (d) affect MOI? POI?
3. Does the location with respect to the axes affect POI? How?
4. If the area is more in quadrant I or II or III or IV—how will each affect POI?
5. What if an area is symmetric, how does that affect POI?
6. What do values of POI mean? What if POI for a shape (and for a set of axes is zero)?
7. Understand each term of the POI PAT: $I_{xy} = \bar{I}_{x'y'} + Adxdy$
8. What are possible values of $\bar{I}_{x'y'}$? Of dx and/or dy?

9. For the shape at right, what are $\bar{I}_{x'y'}$ and A and dx and dy values for each of the component shapes?



Shape	$\bar{I}_{x'y'}$	A	dx	dy
Rectangle				
Circle				
Triangle				